

Influence of Future Predictions on Active Learning

Task 1: Guided Active Learning Method

Our method enhances standard active learning by showing users how their ratings will influence future recommendations. When rating a movie, users see a list of similar movies that will be most affected by their rating. This provides strategic insight into how their input shapes the system's understanding of their preferences.

Task 2: User Study Design

Hypothesis: Providing impact predictions improves rating quality and system accuracy

Participants: 100 users recruited via Prolific Academic

Design: Between-subjects (with/without impact predictions)

Procedure:

1. Pre-study questionnaire (demographics, movie preferences)
2. Rate 20 movies with our interface
3. Evaluate 10 recommendations
4. Post-study interview

Metrics: Recommendation accuracy, rating time, user satisfaction